

Medical Image Analysis

Active Shape Models

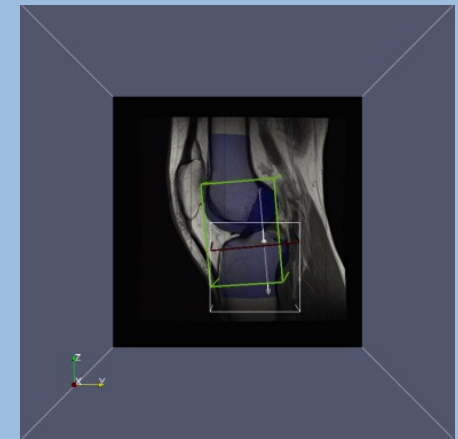
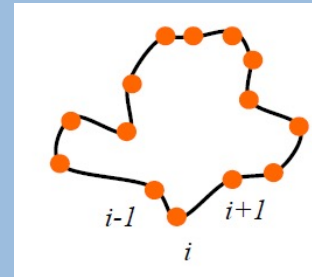
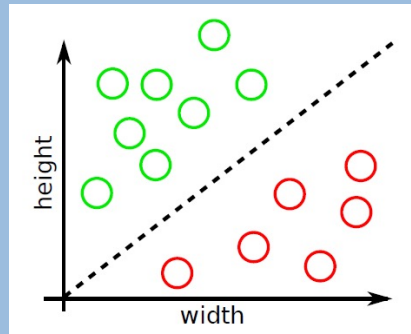
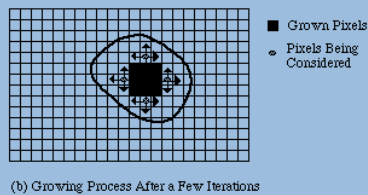
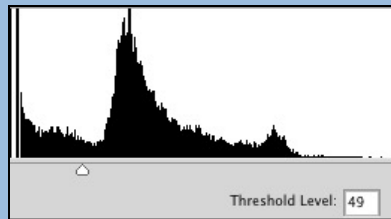
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Lecture Overview

- ◆ Statistical Shape Modeling
 - ◆ Quick summary/review
- ◆ Further considerations
 - ◆ Reference selection
 - ◆ Modeling scaling
 - ◆ Outlier detection
 - ◆ Modeling Physiology
- ◆ Statistical Shape Modeling for Segmentation: Active Shape Models

Segmentation Overview – Big Picture



Simple Methods

- > Thresholding
- > Region-Growing
- > ...

Classification + Clustering

- > kNN
- > SVM
- > ...

Deformable Models

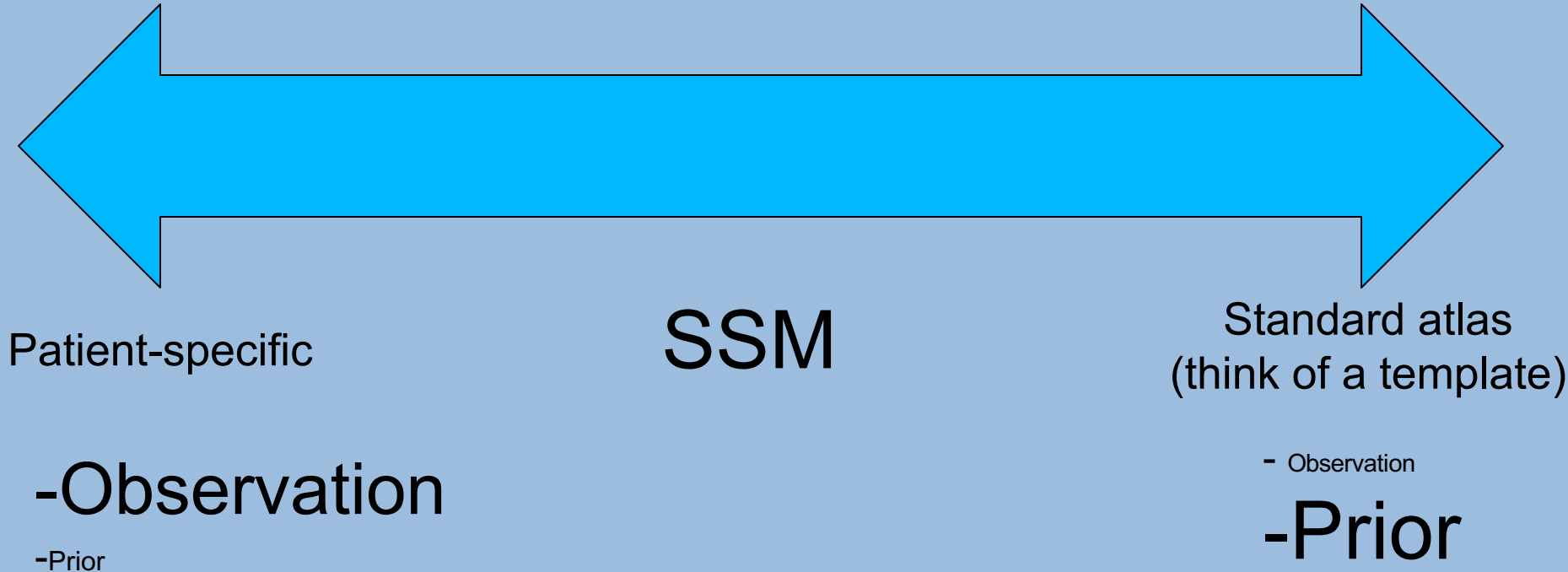
- > Snakes
- > Level-Sets
- > ...

Active Models

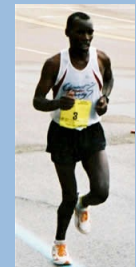
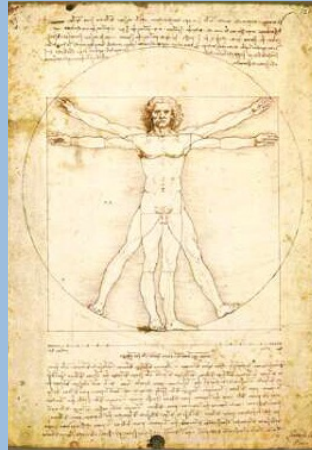
- > **ASM**
- > **AAM**

Motivation Behind SSM of Human Organs

Prior knowledge (what we know in advance)
vs
Observation (what we see)



Statistical distribution of shape



How?!

Mathematical Tools

Principal Component Analysis

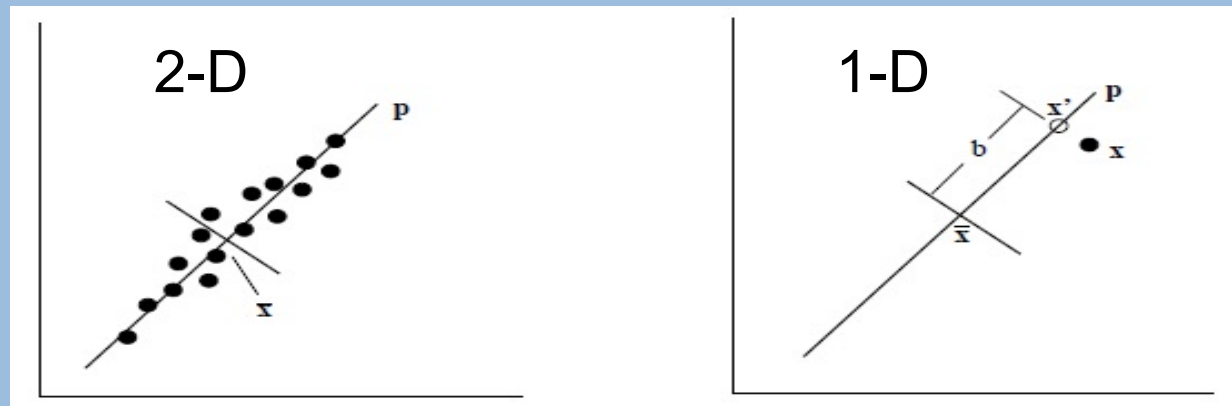
- ◆ Principal component analysis (PCA) is a method used to **reduce the dimensionality** of a linear system
- ◆ PCA projects the original data onto a lower-dimensional space
- ◆ PCA embeds the data in a more compact representation

Let us first define PCA as a mathematical tool, and then see how we can apply it to SSM of anatomical structures

Mathematical Tools

Principal Component Analysis

- Assume that you have 2-D data points
- In this example, data points are scattered along a principal direction
- PCA yields the main direction (and secondary directions), and creates an orthogonal coordinate system that **maximizes the variance** of the points in each direction
- Points can be referenced along the main component (axis) using a parameter b
- How?



Mathematical Tools

Principal Component Analysis

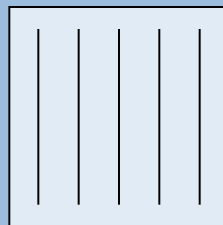
- SVD diagonalizes the covariance matrix of the original data
- It directly yields the eigen-vectors and eigen-values of $Cov(M)$

$$M = U W V^T$$

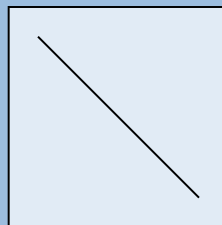
Eigen-vectors

Diagonal matrix of eigen-values

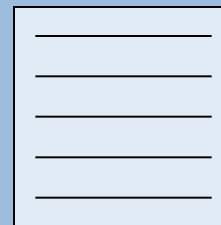
$M =$



U



W



V^T

Mathematical Tools

Principal Component Analysis

- ◆ PCA assumes that the original data follows a Gaussian distribution (even if in reality it does not)
- ◆ PCA results in an **orthogonal** lower-dimensional system. i.e., the principal axes are perpendicular
- ◆ PCA is a **linear** method
- ◆ There also exist several linear and non-linear dimensionality reduction techniques such as factor analysis Isomap, multidimensional scaling (MDS), etc.

Shape Representation

Point Clouds

- Every surface point: $p_i = \{x_i, y_i, z_i\}$
- The surface is represented by concatenating the points

$$S = \{p_1, p_2, \dots, p_n\}$$

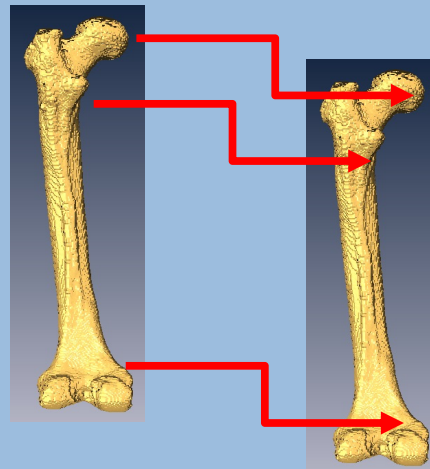
$$\Rightarrow S_i = \{x_{i,1}, y_{i,1}, z_{i,1}, x_{i,2}, y_{i,2}, z_{i,2}, \dots, x_{i,n}, y_{i,n}, z_{i,n}\}$$

- n = number of points that constitute the surface

Point-to-Point Correspondence

- Now assume we have two different objects of the same class (i.e., two femurs from two different patients)
- Initially, isosurfacing results in point clouds with a different number of points
- We should establish anatomical correspondence between the points

S1 S2



Point Distribution Models (PDM)

- So far, we have:
 - A set of aligned shapes
 - Each shape has the same number of surface points
 - Surface points correspond anatomically and/or spatially
 - Each shape is represented by a vector of coordinates
 $s_i = \{x_{i,1}, y_{i,1}, z_{i,1}, x_{i,2}, y_{i,2}, z_{i,2}, \dots, x_{i,n}, y_{i,n}, z_{i,n}\} \in \mathbb{R}^{3n}$
- We can represent the set of shapes in a matrix by concatenating the m shape vectors s_i

$$M = \left(\begin{array}{|c|c|c|} \hline S1 & S2 & S3 \\ \hline \end{array} \quad \begin{array}{|c|} \hline S_m \\ \hline \end{array} \right)$$

Modeling of Shape Variability

- ◆ Subtract mean from each instance of $M \rightarrow D = M - \text{mean}(M)$
- ◆ Apply SVD directly on matrix D

$$\mathbf{D} = \begin{pmatrix} X_{1,1} & X_{2,1} & \cdots & X_{m,1} \\ \vdots & \vdots & \ddots & \vdots \\ X_{1,n} & X_{2,n} & \cdots & X_{m,n} \\ Y_{1,1} & Y_{2,1} & \cdots & Y_{m,1} \\ \vdots & \vdots & \ddots & \vdots \\ Y_{1,n} & Y_{2,n} & \cdots & Y_{m,n} \\ Z_{1,1} & Z_{2,1} & \cdots & Z_{m,1} \\ \vdots & \vdots & \ddots & \vdots \\ Z_{1,n} & Z_{2,n} & \cdots & Z_{m,n} \end{pmatrix} = \mathbf{U} \mathbf{W} \mathbf{V}^T$$

$$\mathbf{D} = \mathbf{U} \mathbf{W} \mathbf{V}^T$$

Eigen-vectors $\rightarrow$ $\mathbf{U}$
 Diagonal matrix of eigen-values $\rightarrow$ $\mathbf{W}$

$$\mathbf{D} = \begin{array}{|c|} \hline \text{Vertical lines} \\ \hline \end{array} \begin{array}{|c|} \hline \text{Diagonal line} \\ \hline \end{array} \begin{array}{|c|} \hline \text{Horizontal lines} \\ \hline \end{array}$$

$\mathbf{U} \quad \mathbf{W} \quad \mathbf{V}^T$

$\mathbf{U}$ = eigen-vectors of $\text{Cov}(M)$
 $\text{diag}(\mathbf{W})$ = eigen-values of $\text{Cov}(M)$

n = number of points per mesh
 m = number of training samples

Summary Statistical Shape Modelling

Model $\bar{x} + b \cdot \phi$ $\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i$ Mean Shape

Statistical Analysis via Principal Component Analysis (PCA)

Eigen-decomposition of matrix

$$S = \frac{1}{N} \sum_{i=1}^N dx_i dx_i^T \quad dx_i = x_i - \bar{x}$$

provides eigenvectors/eigenvalues (λ, ϕ)

$$S \phi_k = \phi_k \lambda_k \quad \text{With } k, \text{ the } k\text{th eigen-vector/value}$$

Model Evaluation

- ◆ Commonly, three measures are used to evaluate a statistical shape model, namely
 - ◆ Compactness
 - ◆ *Generalization*
 - ◆ Specificity
- ◆ They are used to indicate how well the modeling process has been able to embed the original data into a lower dimensional space

Model Evaluation

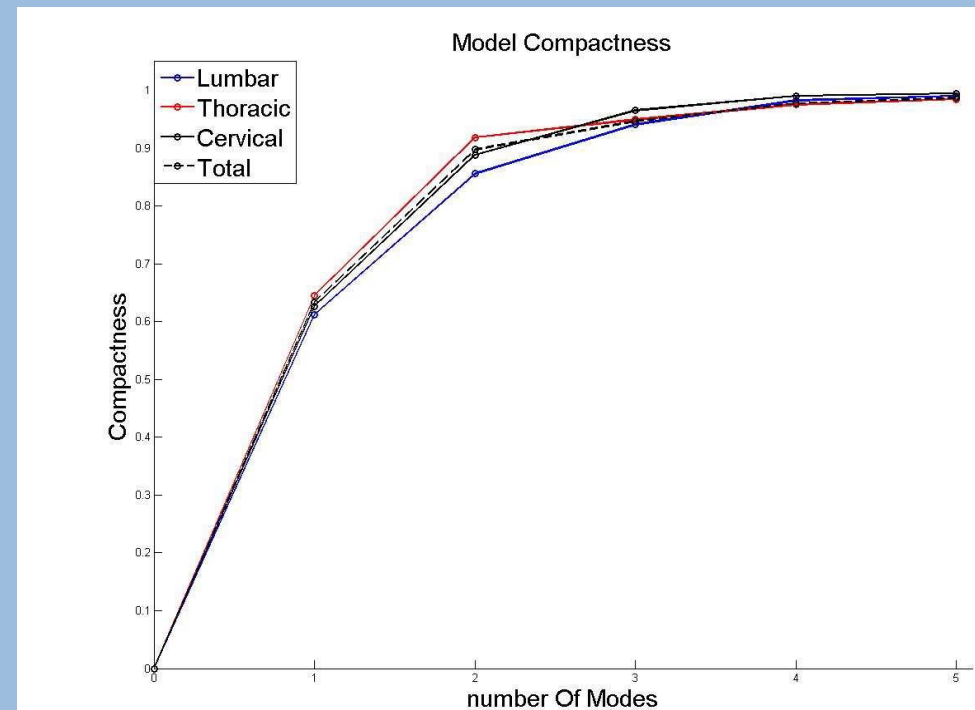
Compactness

- ◆ Compactness is a measure of how good the data reduction process is.
- ◆ In other words, a model that is compact enough, allows the generation of new shape instances using as few parameters (principal modes or components) as possible
- ◆ It is represented by a curve showing how much of the total variance a certain number of modes of variation can capture

Model Evaluation Compactness

- ♦ t = number of modes used
- ♦ λ_i = eigen-values
- ♦ λ_{total} = total variance = all eigen-values summed together

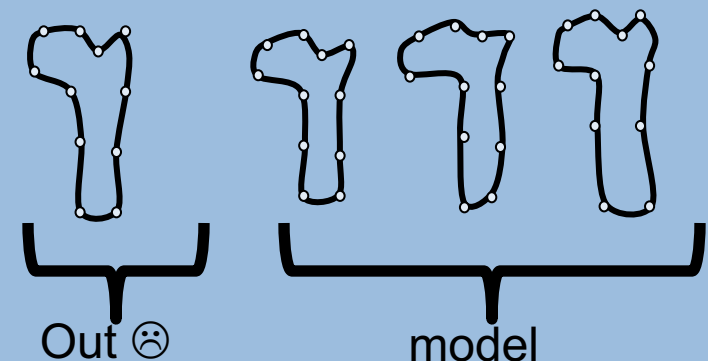
$$C(\lambda_t) = \frac{\lambda_t}{\lambda_{\text{total}}}$$



Model Evaluation Generalization

- Generalization is performed using **leave-one-out (cross-validation)** reconstruction experiments
- It represents the ability of the model to generate new instances from the class
- For every left-out instance, a new statistical model is computed from the remaining training instances
- The parameters of the left-out instance are computed

$$b_{out} = \phi^{-1}(x_{out} - \bar{x})$$



Model Evaluation Generalization

- ♦ The parameters are used to reconstruct the instance

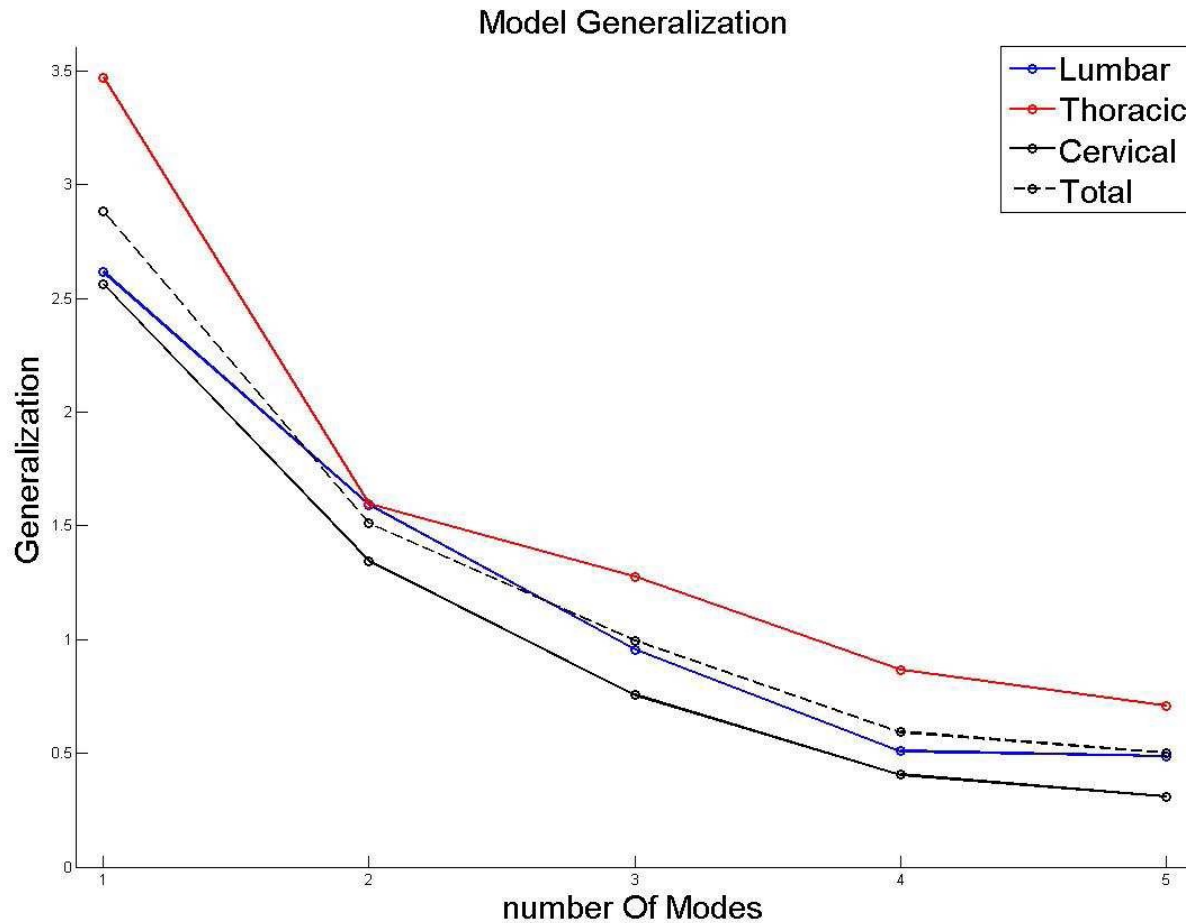
$$\tilde{x} = \bar{x} + \phi b_{out}$$

- ♦ The errors are measured and averaged over all training instances yielding the **generalization curve** of the model generalization

$$\frac{1}{N} \sum_i (x_i - \tilde{x}_i)^2 \quad \text{i: sample reconstructed}$$

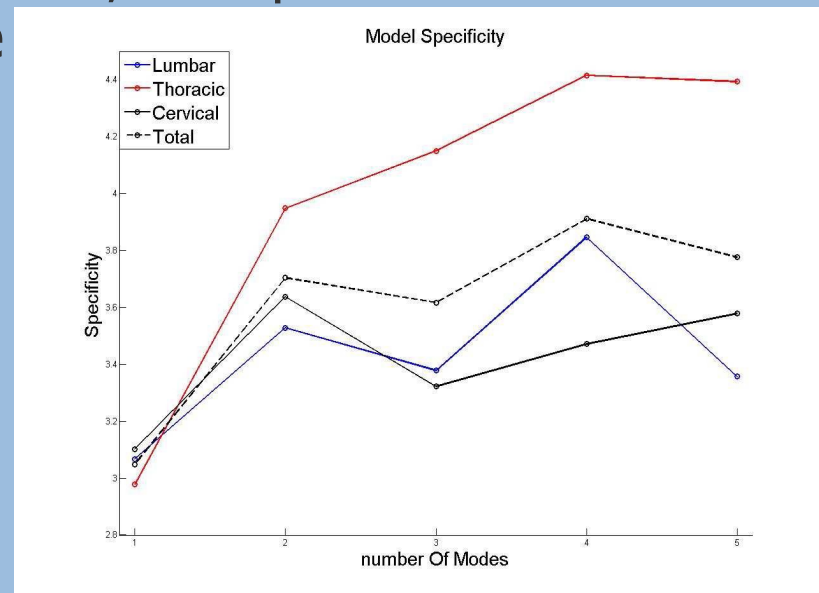
- ♦ It is also dependent on the number of modes of variation (parameters used in the reconstruction process)

Model Evaluation Generalization



Model Evaluation Specificity

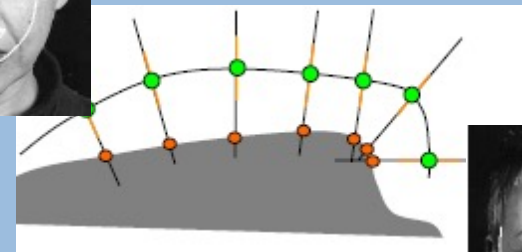
- Specificity is an indicator of **how good the model is in generating instances similar to those presented in the training set**
- It is measured by generating a large number of random instances using different number of modes, and for every new instance, compute the distance to the closest shape in the



SSM for Segmentation: Active Shape Models

Active Shape Models – ASM

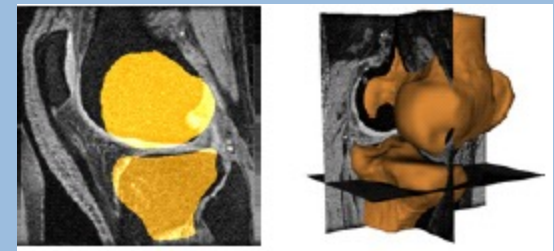
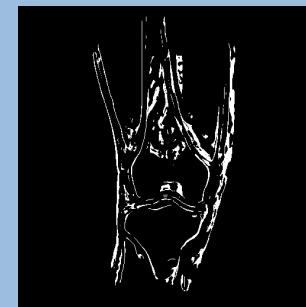
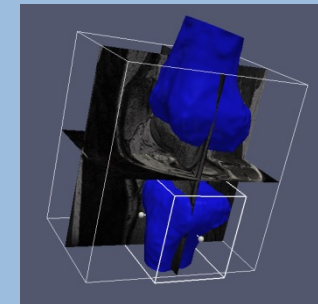
- > General idea
 - Start with SSM and image
 - Initially place SSM in image
 - Sample along SSM surface normals, find largest gradient (= edge)
 - Deform pose (scale, rotation, translation) and shape (PCA modes) of model to best fit to extracted image features (largest gradient)



Cootes et al. 95'

Initialization – A Crucial Step

- > Approximately align SSM with structure of interest
- > Many different methods, choose the one for your application
 - User interaction
 - Centroid alignment
 - Chamfer matching
 - Atlas registration
 - Global search on entire image (evolutionary algorithms)



Initialization – A Crucial Step

Importance of Initialization



Initial



2 Iterations



20 Iterations

Feature Extraction

> Edge detection methods

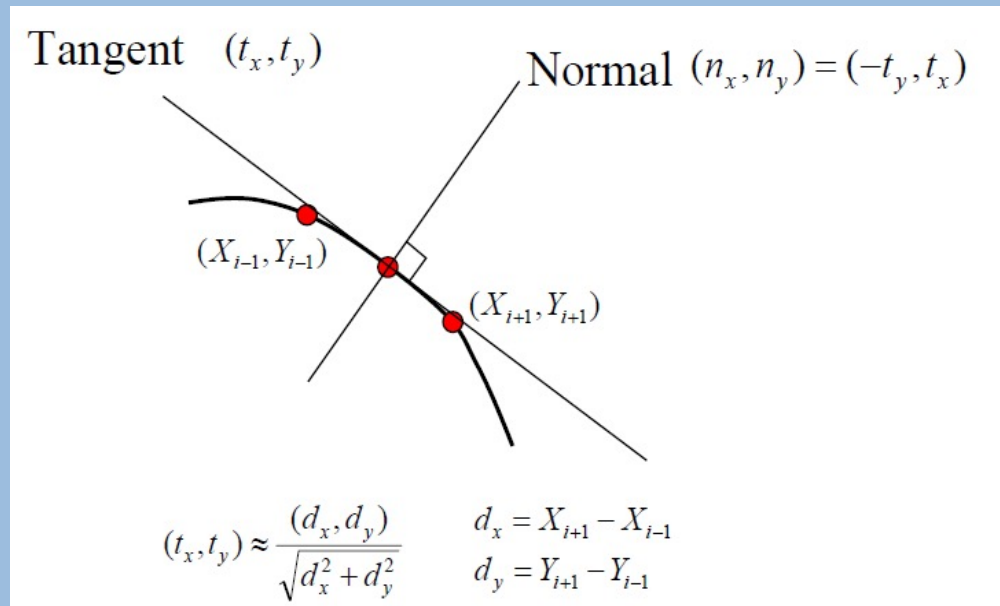
- _____
- _____
-

$$\mathbf{G}_y = \begin{bmatrix} +1 & +2 & +1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix} * \mathbf{A} \quad \text{and} \quad \mathbf{G}_x = \begin{bmatrix} +1 & 0 & -1 \\ +2 & 0 & -2 \\ +1 & 0 & -1 \end{bmatrix} * \mathbf{A}$$



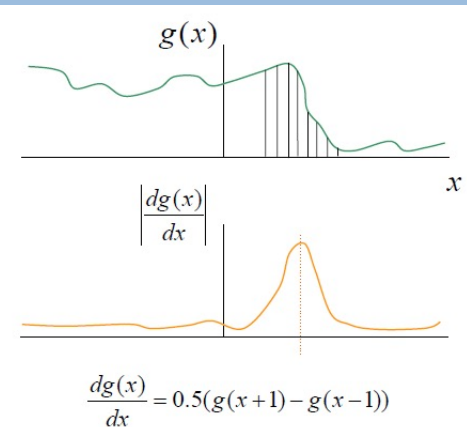
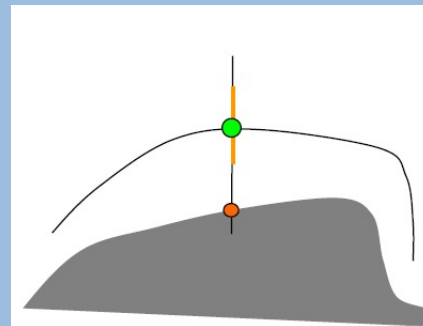
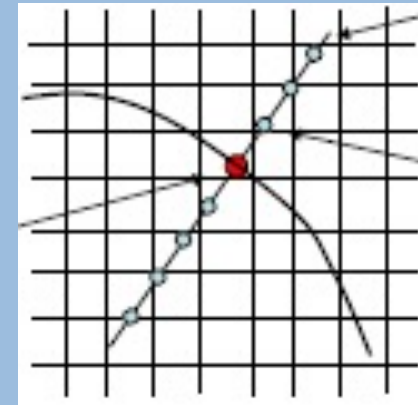
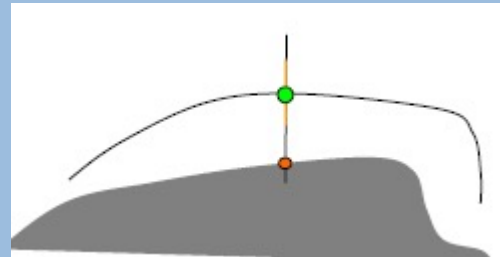
Normals

- > SSM
 - points + connections give mesh
 - So we know the neighbors of each point
 - _____ of two vectors yields orthogonal vector
 - This is the surface normal



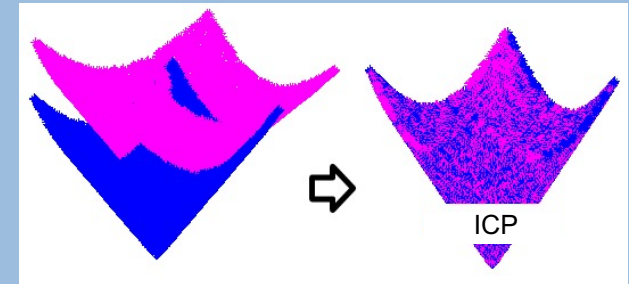
Sampling along profiles

- > Given:
 - Point on SSM
 - Normal at that point
- > Sample along normal profile
- > Parameters
 - Sampling range
 - Sampling step size
 - Yields $2k+1$ samples
- > Choose point with largest feature value (gradient)
- > Model point should be transformed to that point



Pose Estimation

- > Goal: Rough alignment of both shapes
- > Here: similarity transform in 2D
- > Optimize registration process according to certain cost function (e.g. mean Euclidean distance)



$$\bar{X} = T_{X_t, s, \Theta}(\bar{x})$$

$$T_{X_t, s, \Theta} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x_t \\ y_t \end{pmatrix} + \begin{pmatrix} s \cos \Theta & -s \sin \Theta \\ s \sin \Theta & s \cos \Theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\text{minimize } \|Y - X\| = \|Y - T_{X_t, s, \Theta}(\bar{x})\|$$

Shape Estimation

- > PCA-based deformation
- > Keep only k most important PCA modes (e.g. so that k modes account for 90% of variance)
- > Threshold b to allow for plausible shapes only, e.g. $|b_i| < 3 \sigma_i$ (3 std.dev. capture more than 95% of variation in Gaussian distribution)

$$x \approx \bar{x} + Pb$$

$$b = P^T (x - \bar{x})$$

$$b = P^T (Y - \bar{x})$$

Optimization Schemes

- > Optimize pose and shape part separately or jointly
- > Optimization
 - Choose optimizer for pose estimation (e.g. gradient descent)
 - Closed form solution for shape
- > Iterative method:
 - Compute profile along normals
 - Adjust pose
 - Adjust shape
 - Iterate from beginning until convergence

$$X = T_{X_t, s, \Theta}(\bar{x} + Pb)$$

$$Y = T_{X_t, s, \Theta}(\bar{x} + Pb)$$

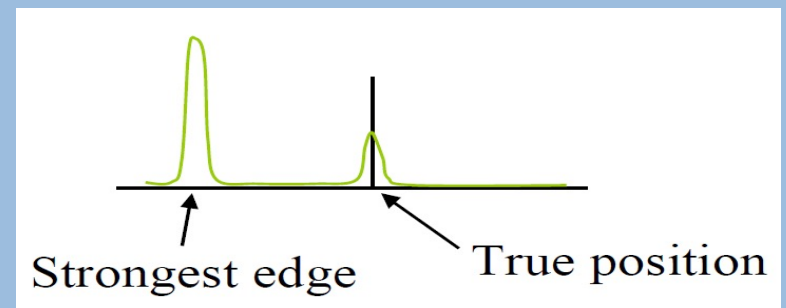
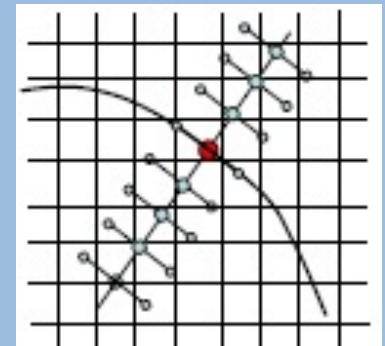
$$b = P^T (T_{X_t, s, \Theta}^{-1}(Y) - \bar{x})$$

Summary – The Complete ASM Algorithm

1. Examine a region of the image around each point $\mathbf{X}_i$ to find the best nearby match for the point $\mathbf{X}'_i$
2. Update the parameters $(X_t, Y_t, s, \theta, \mathbf{b})$ to best fit the new found points $\mathbf{X}$
3. Apply constraints to the parameters, $\mathbf{b}$, to ensure plausible shapes (eg limit so $|b_i| < 3\sqrt{\lambda_i}$).
4. Repeat until convergence.

Advanced Methods

- > Multiresolution pyramid
- > Handling noise by averaging along normal points
- > Intensity profiles



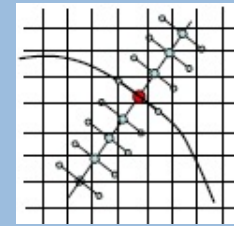
Advanced Methods

> Intensity profiles: Modeling local structure

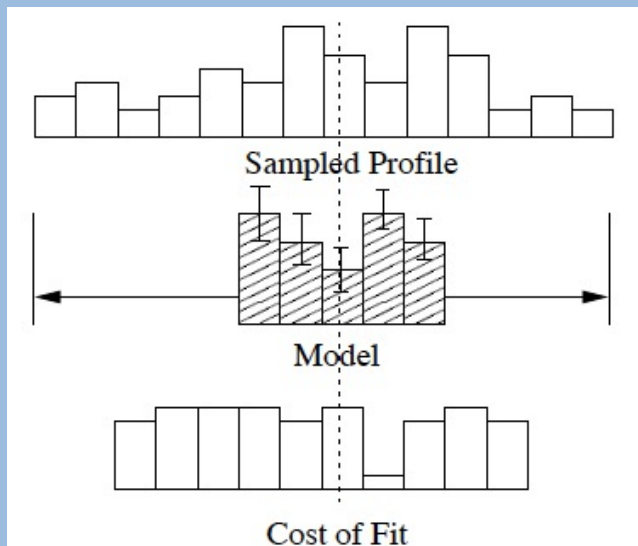
1. Samples along profile for ith image
2. Avoid intensity changes-> use gradient values
3. Normalize
4. Repeat for each data point
5. Assume Gaussian distribution: $(\bar{g}, S_g)$
6. Quality of fit for new sample s: Mahalanobis distance

$$f(\mathbf{g}_s) = (\mathbf{g}_s - \bar{\mathbf{g}})^T \mathbf{S}_g^{-1} (\mathbf{g}_s - \bar{\mathbf{g}})$$

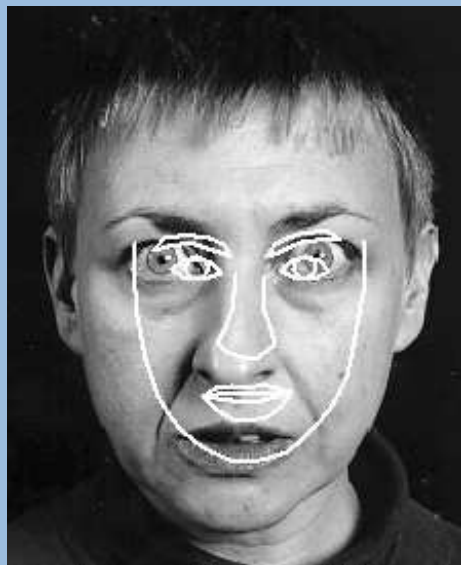
7. Pick lowest values (higher probability)



$$\mathbf{g}_i \rightarrow \frac{1}{\sum_j |g_{ij}|} \mathbf{g}_i$$



Examples – ASM 2D



Initial



2 Iterations

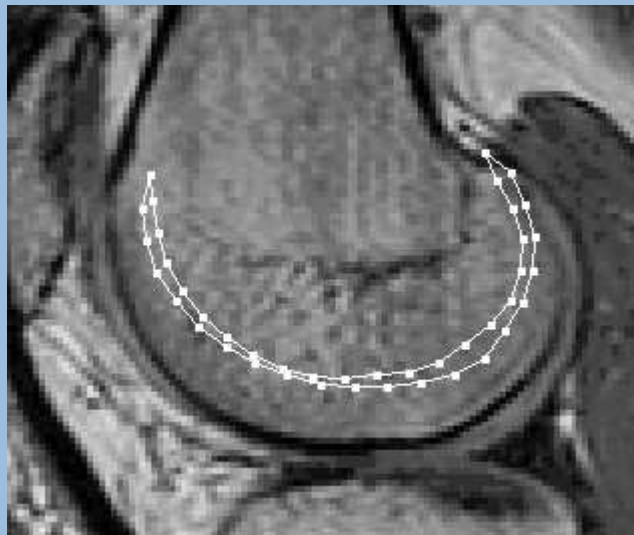


6 Iterations

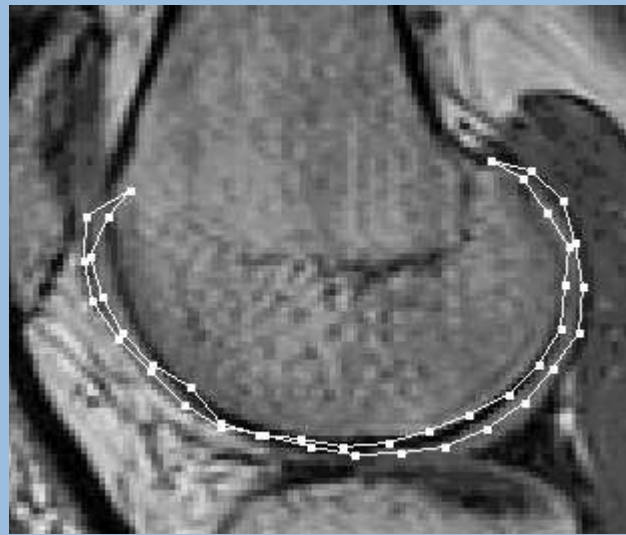


18 Iterations

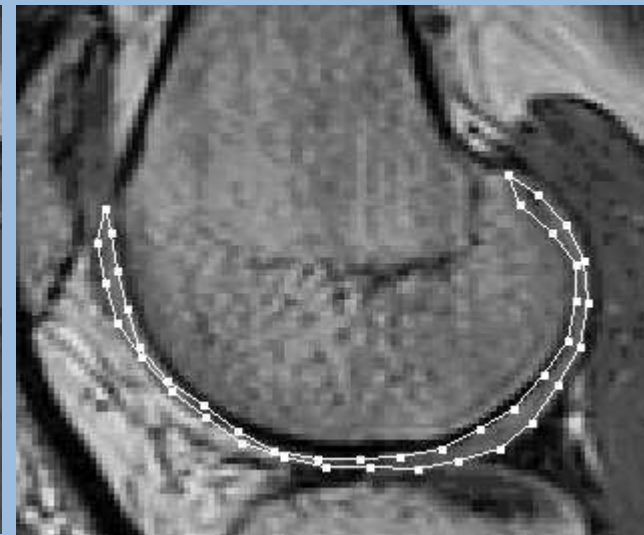
Clinical example



Initial

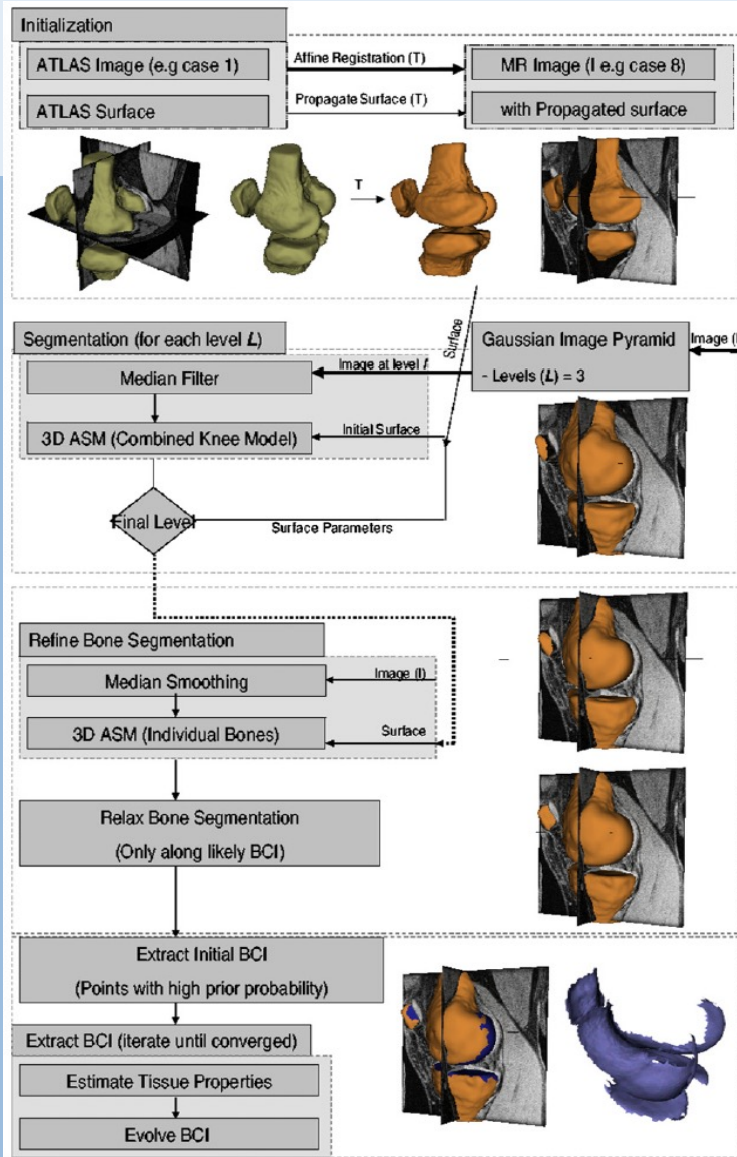


1 Iteration

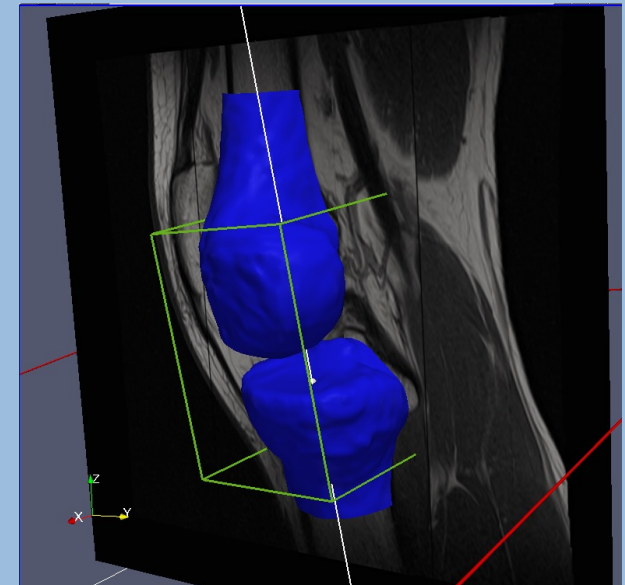
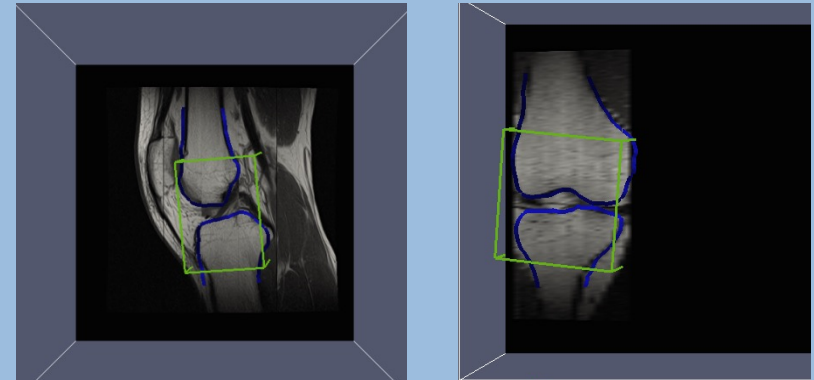


14 Iterations

Examples – ASM 3D

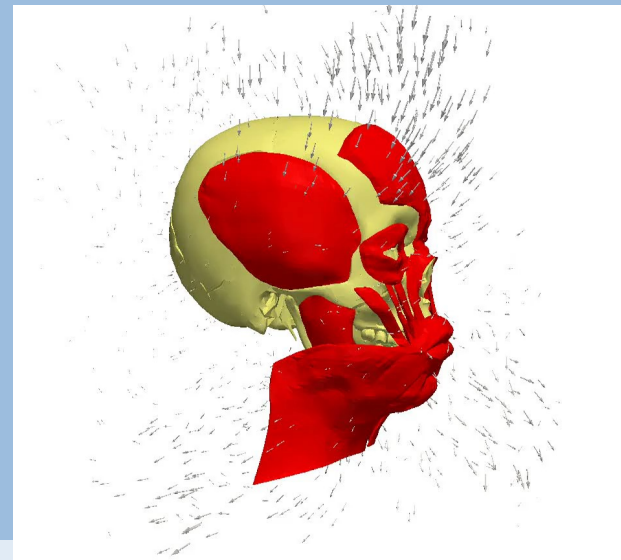
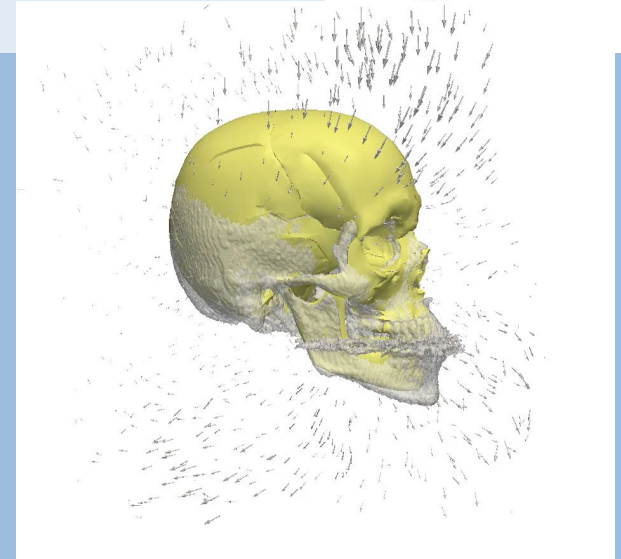
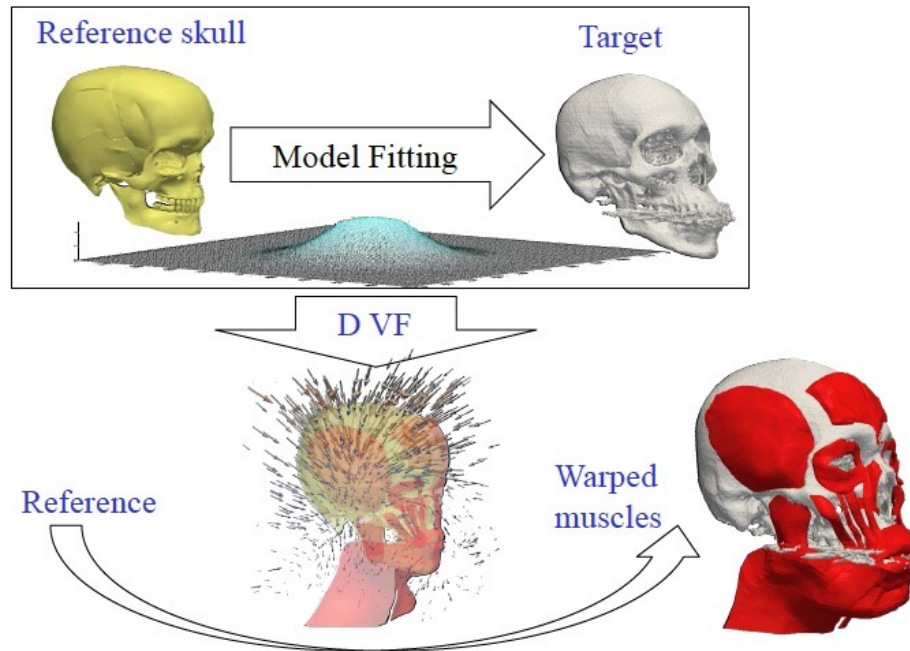


[Fripp 2007]



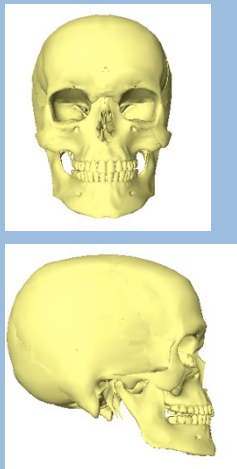
[Bauer 2012]

Example in CMF

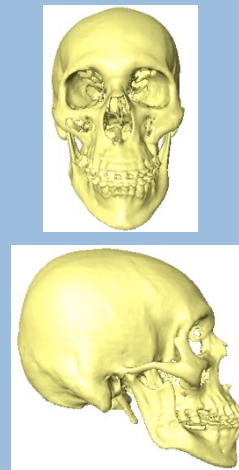


Example in CMF

ASM enhanced with muscle information



Model

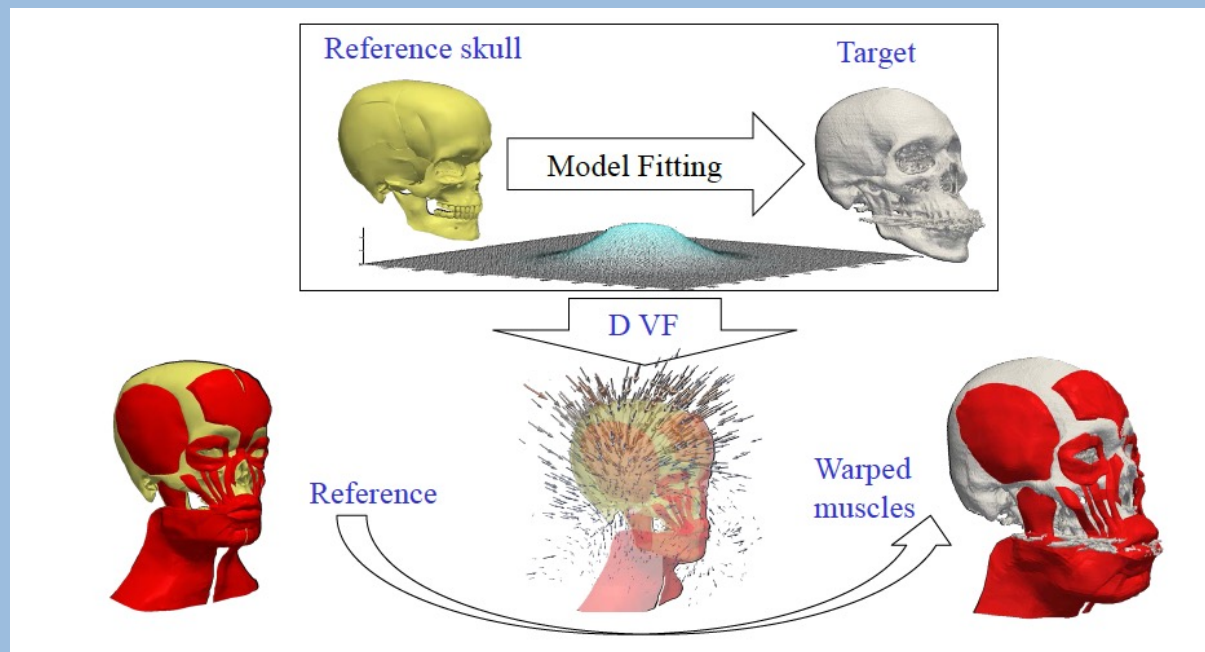


Patient

Example in CMF

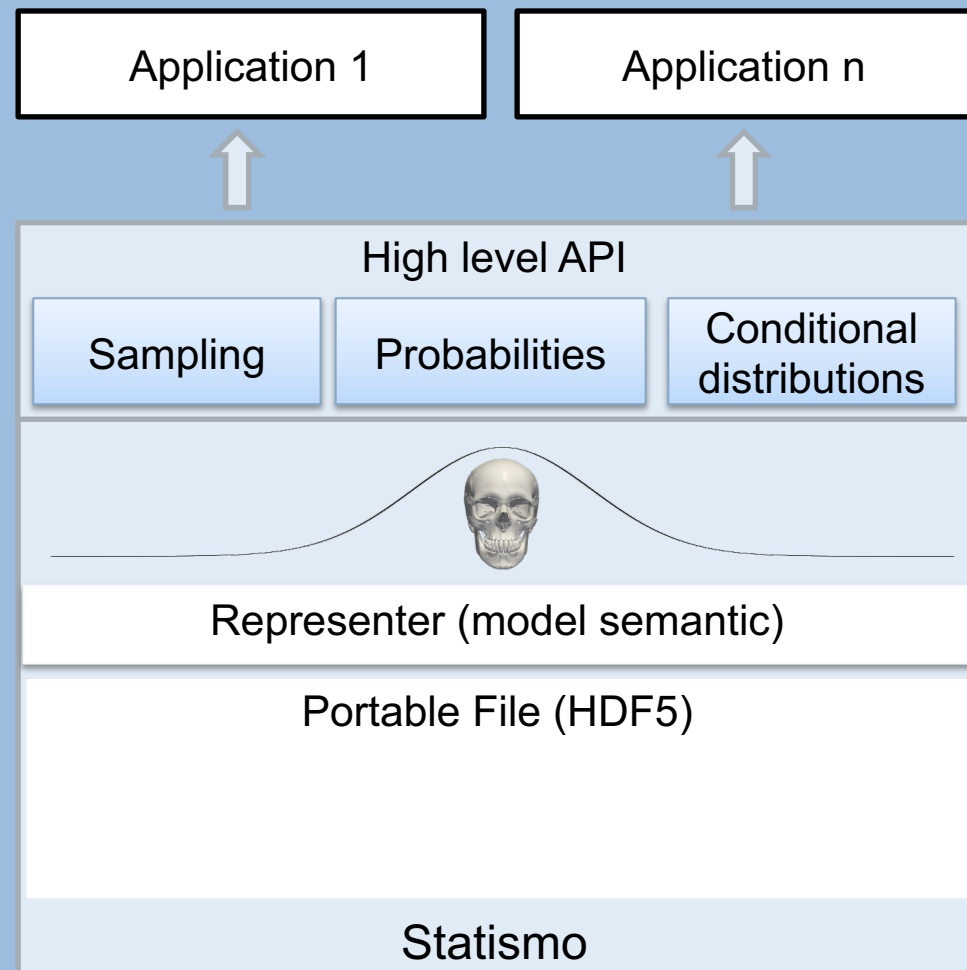
Model enhanced with:

- Facial muscles
- Muscle fibers (anisotropy of tissue properties)
- Predefined surgical approaches
- Data preprocessing (e.g. cropping)
- Simulation-specific (e.g. sliding contact areas)



Statismo

- > Statismo provides a high level API.
- > Users work with shapes, not vectors.
- > Core functionality implemented only once.
- > Exchanging models becomes easy.



Statismo is Open Source

Get it at
<http://statismo.github.com>

Statismo - A framework for PCA based statistical models

*Lüthi M., Blanc R., Albrecht T., Gass T., Goksel O., Büchler P., Kistler M., Bousleiman H.,
Reyes M., Cattin P., Vetter T.*

Insight Journal, <http://hdl.handle.net/10380/3371>

Building texture models

- > Warp each training sample to mean shape to obtain mean-free patch
- > Sample intensities from shape-normalized image to form a texture vector
- > Normalize intensity to eliminate *lighting* variations
- > Start from the beginning, same as SSM
 - PCA on normalized texture data

$$g \approx \bar{g} + P_g b_g$$

- > We want to obtain a combined appearance model, which contains both shape and texture variations

Combined models of appearance

- > Shape and texture summarized by parameter vectors b
- > Handle correlations between shape and texture parameters b_s and b_g
- > Apply PCA on combined b to get eigenvectors P_c and eigenvalues c
- > c is vector of appearance parameters controlling both shape and texture

$$b = \begin{pmatrix} W_s b_s \\ b_g \end{pmatrix}$$

Weighting to account for unit differences

$$b = P_c c$$

Combined Models of Appearance

- > Linear model
- > We can express shape and texture directly as function of c
- > How to generate a new image:
 - Generate shape-free intensity image from g
 - Warp it using the control points described by x

$$x = \bar{x} + Q_s c$$

$$g = \bar{g} + Q_g c$$

$$Q_s = P_s W_s^{-1} P_{cs}$$

$$Q_g = P_g P_{cg}$$

Model Fitting

- > Minimize difference between new image and the synthetic AAM image
- > Key step: estimate parameter update from current image sample
- > Difference vector

$$\delta I = I_i - I_m$$



Model Fitting – More Detailed 1

- > Estimate update step using Jacobian
- > Residual difference $r(p)$
- > p is parameter vector $p(c,t,u)$
- > Minimize error $E(p)$
- > Taylor expansion of $r(p+\delta p)$
- > Choose δp to minimize $E(p+\delta p)$
- > Need estimate of Jacobian J
- > Estimate J by small displacements on training set

$$r(p) = g_s - g_m$$

$$E(p) = |r(p)|^2$$

$$r(p + \delta p) = r(p) + \frac{\partial r}{\partial p} \delta p$$

$$E(p + \delta p) = |r(p + \delta p)|^2$$

$$\delta p = -Rr(p)$$

$$R = \left(\frac{\partial r^T}{\partial p} \frac{\partial r}{\partial p} \right)^{-1} \frac{\partial r^T}{\partial p}$$

$$J = \frac{\partial r}{\partial p}$$

Model Fitting – More Detailed 2

- > Iterative algorithm:
 - Measure residual error $r(p)$
 - Predict correction $\delta p = -Rr(p)$
 - $p = p + k \delta p$
 - Repeat to convergence

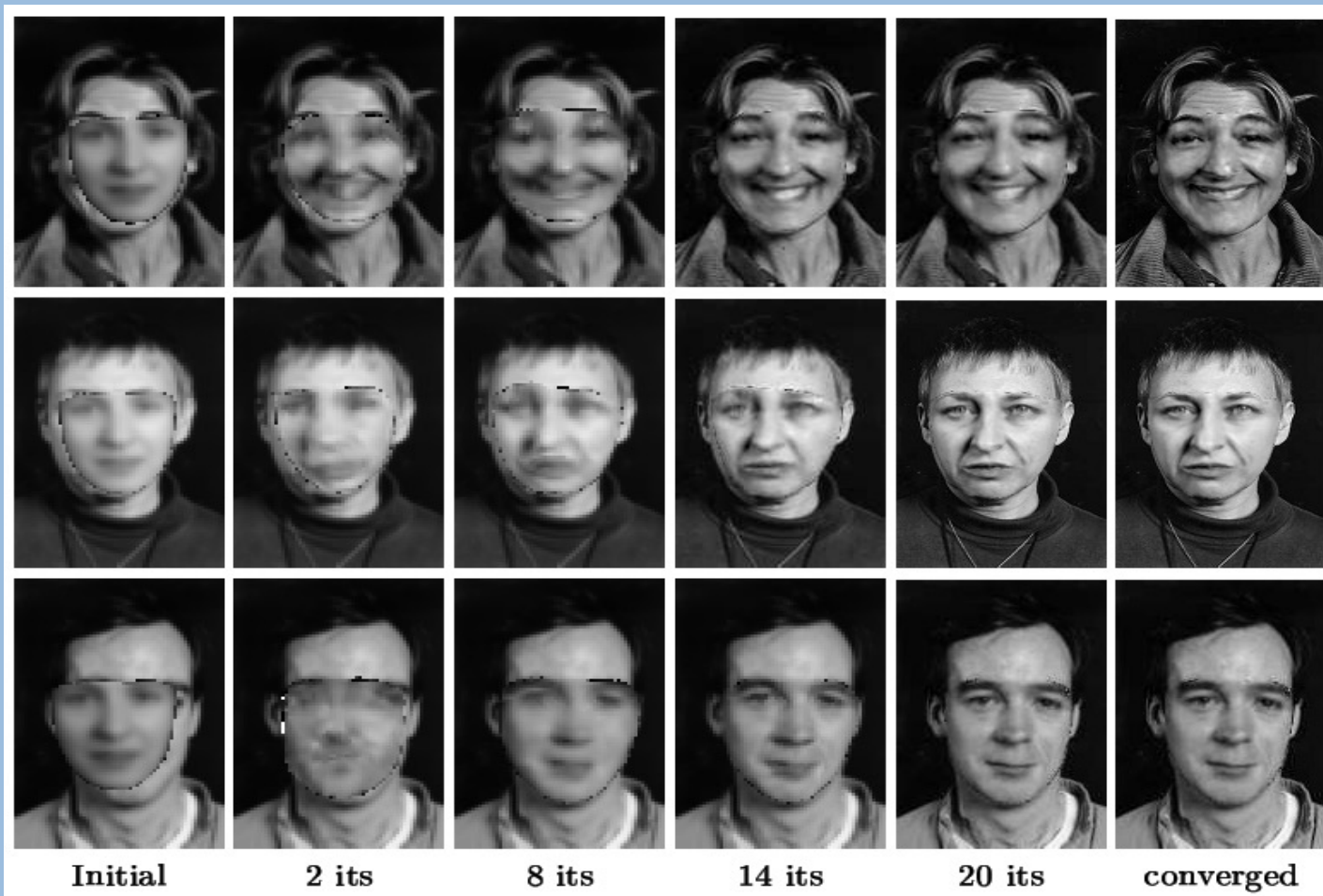
$$r(p) = g_s - g_m$$

$$\delta p = -Rr(p)$$

Summary – The Complete AAM Algorithm

1. Project the texture sample into the texture model frame using $\mathbf{g}_s = T_{\mathbf{u}}^{-1}(\mathbf{g}_{im})$
2. evaluate the error vector, $\mathbf{r} = \mathbf{g}_s - \mathbf{g}_m$, and the current error, $E = |\mathbf{r}|^2$
3. compute the predicted displacements, $\delta \mathbf{p} = -\mathbf{R} \mathbf{r}(\mathbf{p})$
4. update the model parameters $\mathbf{p} \rightarrow \mathbf{p} + k \delta \mathbf{p}$, where initially $k = 1$,
5. calculate the new points, $\mathbf{X}'$ and model frame texture $\mathbf{g}'_m$
6. sample the image at the new points to obtain $\mathbf{g}'_{im}$
7. calculate a new error vector, $\mathbf{r}' = T_{\mathbf{u}'}^{-1}(\mathbf{g}'_{im}) - \mathbf{g}'_m$
8. if $|\mathbf{r}'|^2 < E$ then accept the new estimate, otherwise try at $k = 0.5$, $k = 0.25$ etc.
9. Repeat until convergence

Examples – AAM

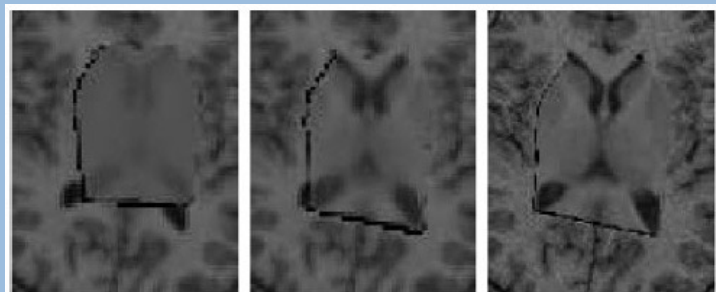


Example: AAM in face synthesis

- > Parameters: Pose, shape, illumination



Examples – AAM



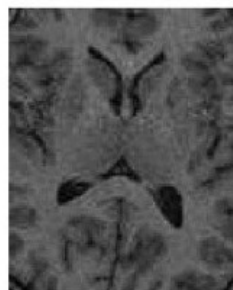
Initial

2 its

6 its

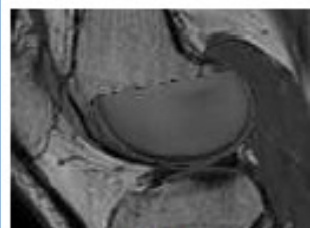


16 its (converged)



original

[Cootes 2004]



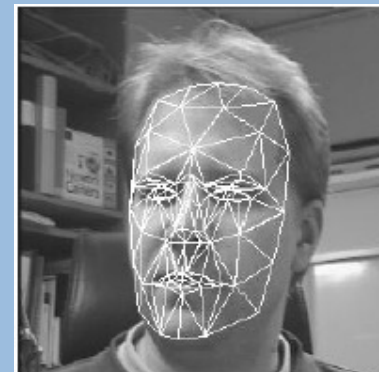
Initial



2 its



Converged (11 its)



[Dornaika 2003]

Comparison ASM – AAM

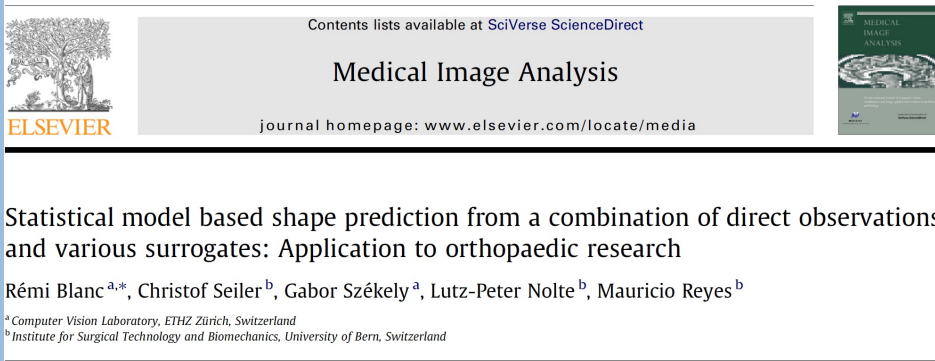
Differences

- > ASM searches around current position along normals, AAM searches at current position
- > ASM minimizes distance between model and corresponding points, AAM minimizes difference between synthetic and target image

Evaluation

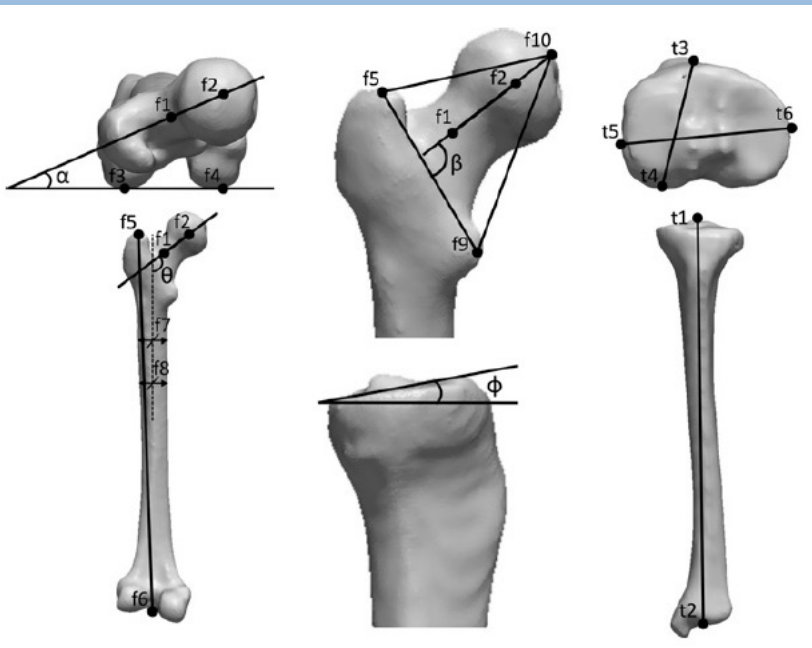
- > ASM usually has larger capture range
- > ASM is faster
- > ASM usually has more accurate feature point location
- > AAM gives better match to image texture

Conditional Shape Models



Aim: From surrogate variables to full shape

Surrogate variables can be length, angles, surface points, etc.



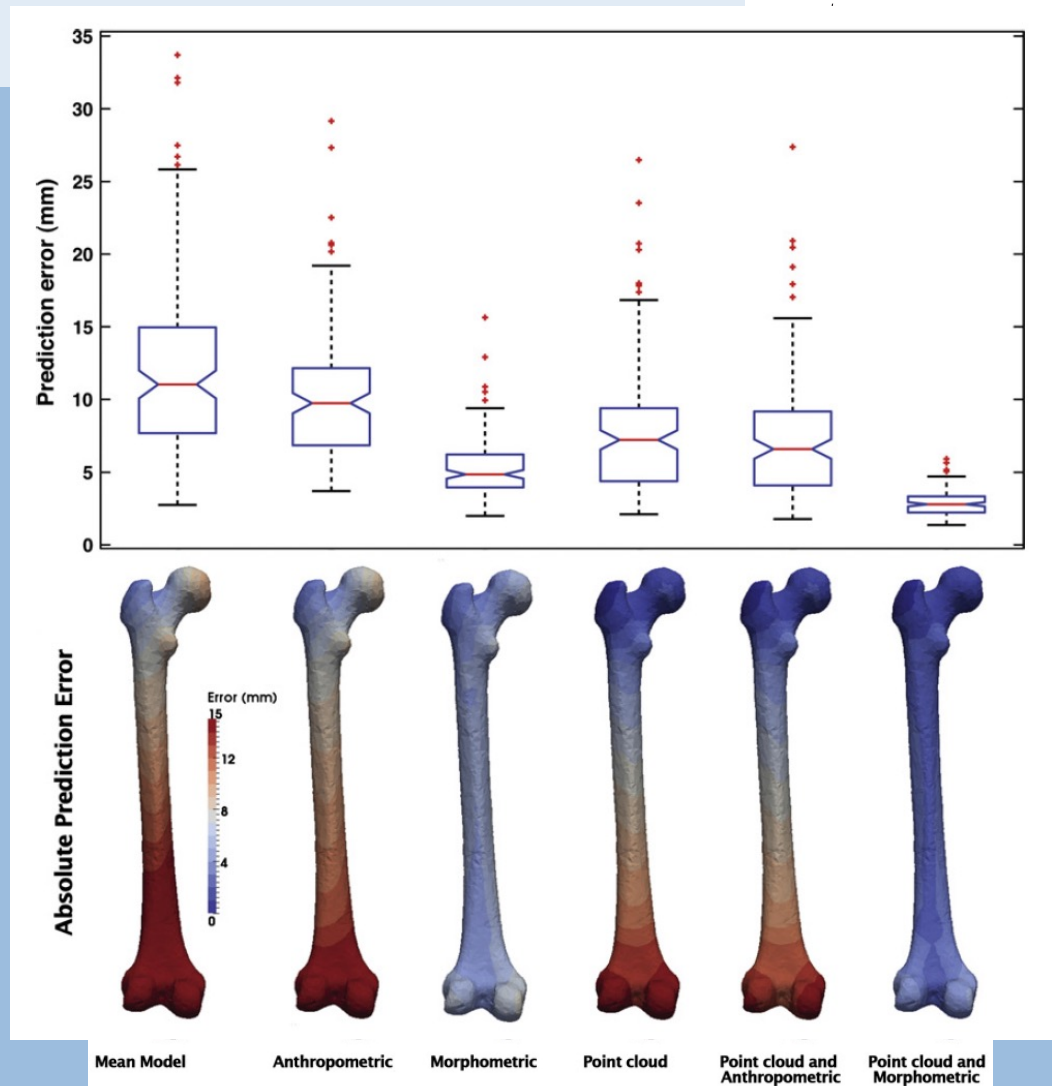
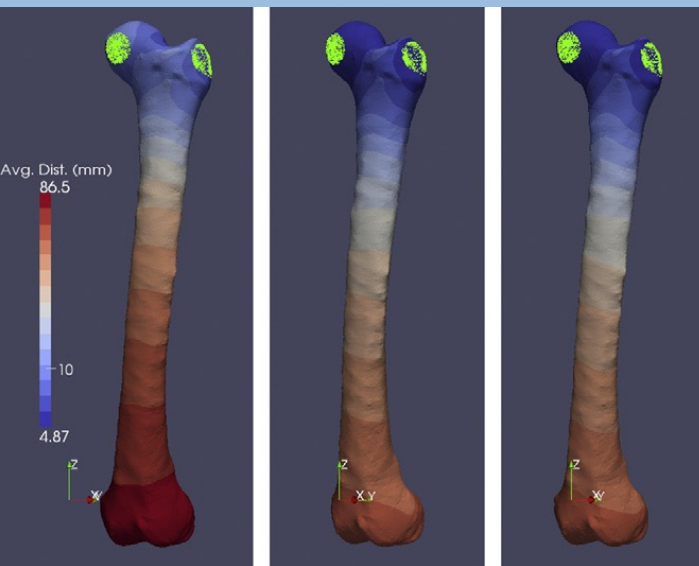
Conditional Shape Models



Statistical model based shape prediction from a combination of direct observations and various surrogates: Application to orthopaedic research

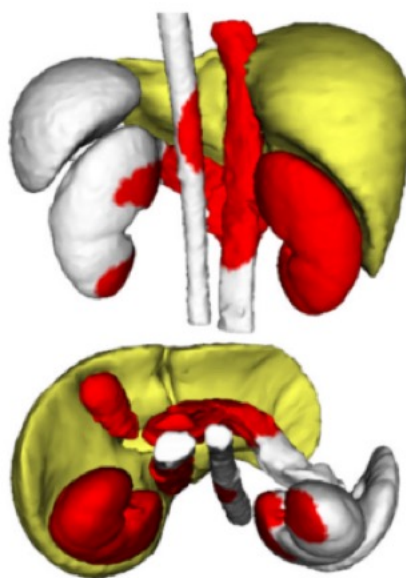
Rémi Blanc^{a,*}, Christof Seiler^b, Gabor Székely^a, Lutz-Peter Nolte^b, Mauricio Reyes^b

^aComputer Vision Laboratory, ETH Zürich, Switzerland
^bInstitute for Surgical Technology and Biomechanics, University of Bern, Switzerland

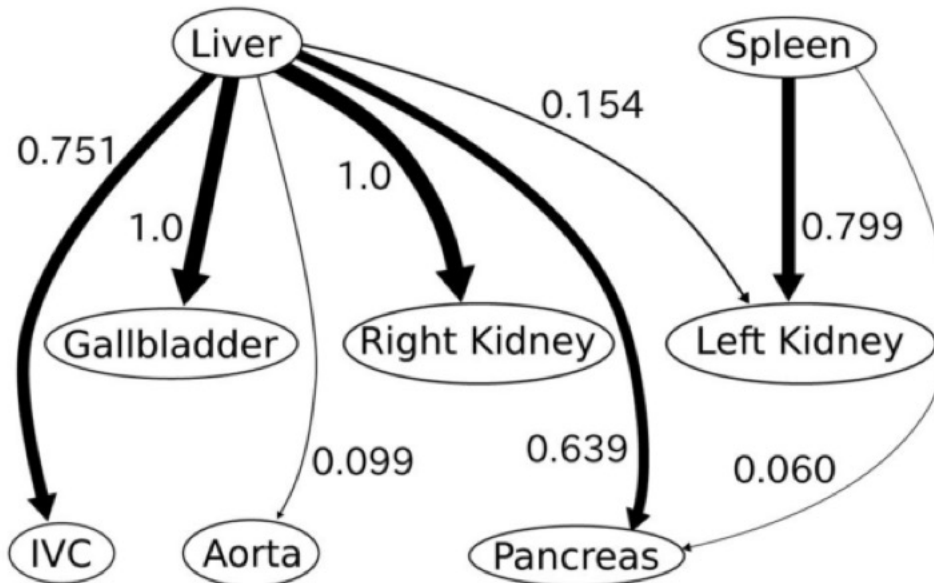


Corrections on MultiOrgan Segmentations :COSMOS

- > Aim: Modify one, correct many!
- > How: Conditional Statistical Shape Models



(a)



(b)

Corrections on MultiOrgan Segmentations :COSMOS

- > Aim: Modify one, correct many!
- > How: Conditional Statistical Shape Models

$$(\sigma = \mathbf{U}\mathbf{D}^2\mathbf{U}^T) \quad \text{Covariance Matrix decomposition}$$

$$\mathbf{s}(\alpha) = \mu + \mathbf{U}\mathbf{D}\alpha = \mu + \mathbf{Q}\alpha. \quad \text{SSM}$$

$$\mathcal{N}(\mu_c, \sigma_c) \quad \text{Posterior shape model given partial or complete corrected organ}$$

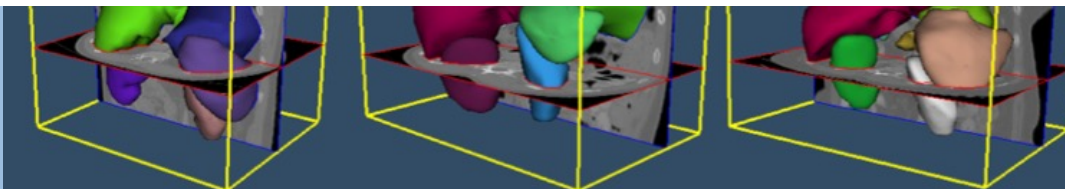
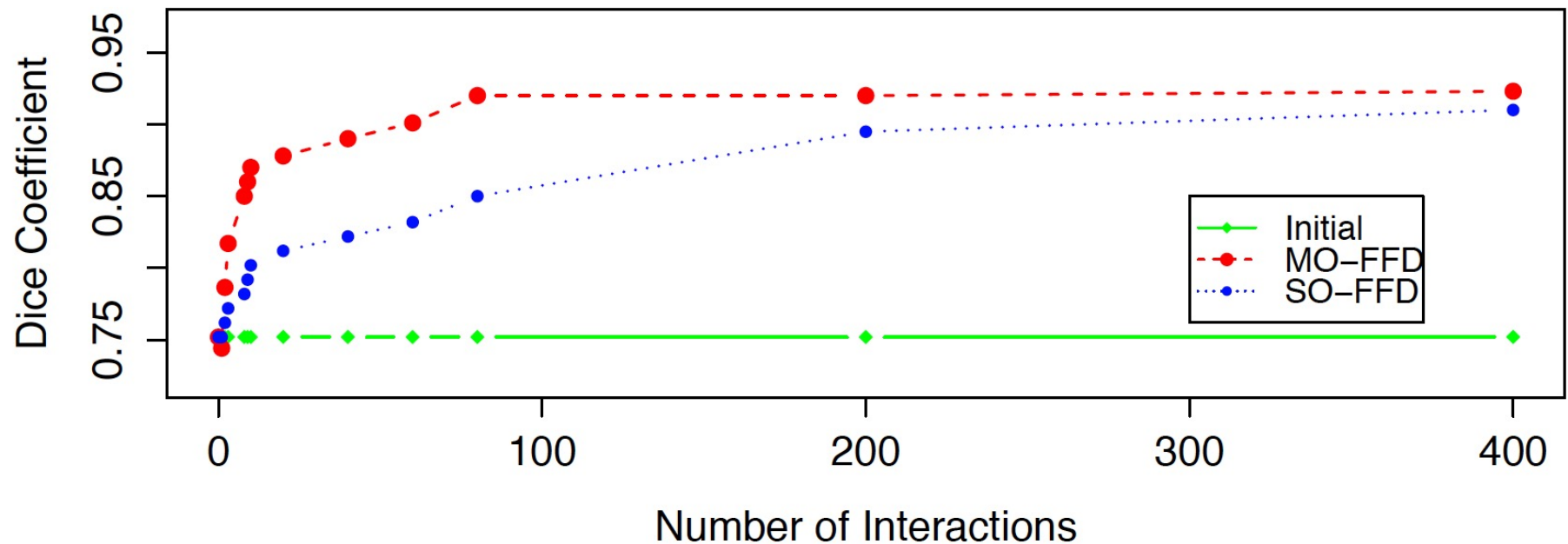
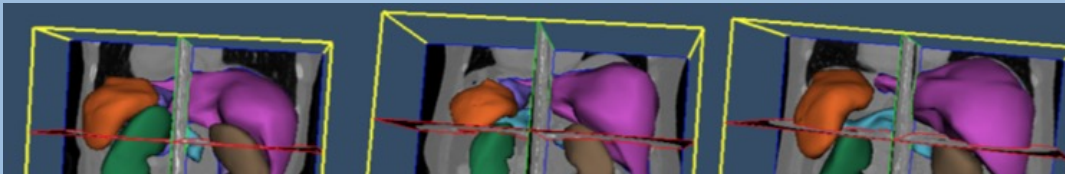
$$p(\mathbf{s}|\mathbf{s}_g) = \mathcal{N}(\mu + \mathbf{Q}\eta, \mathbf{Q}\Lambda\mathbf{Q}^T)$$

$$\mu_c = \mu + \mathbf{Q}(\mathbf{Q}_g^T\mathbf{Q}_g + \sigma^2\mathbf{I}_n)^{-1}\mathbf{Q}_g^T(\mathbf{s}_g - \mu_g)$$

$$\Sigma_c = \sigma^2\mathbf{Q}(\mathbf{Q}_g^T\mathbf{Q}_g + \sigma^2\mathbf{I}_n)^{-1}\mathbf{Q}^T,$$

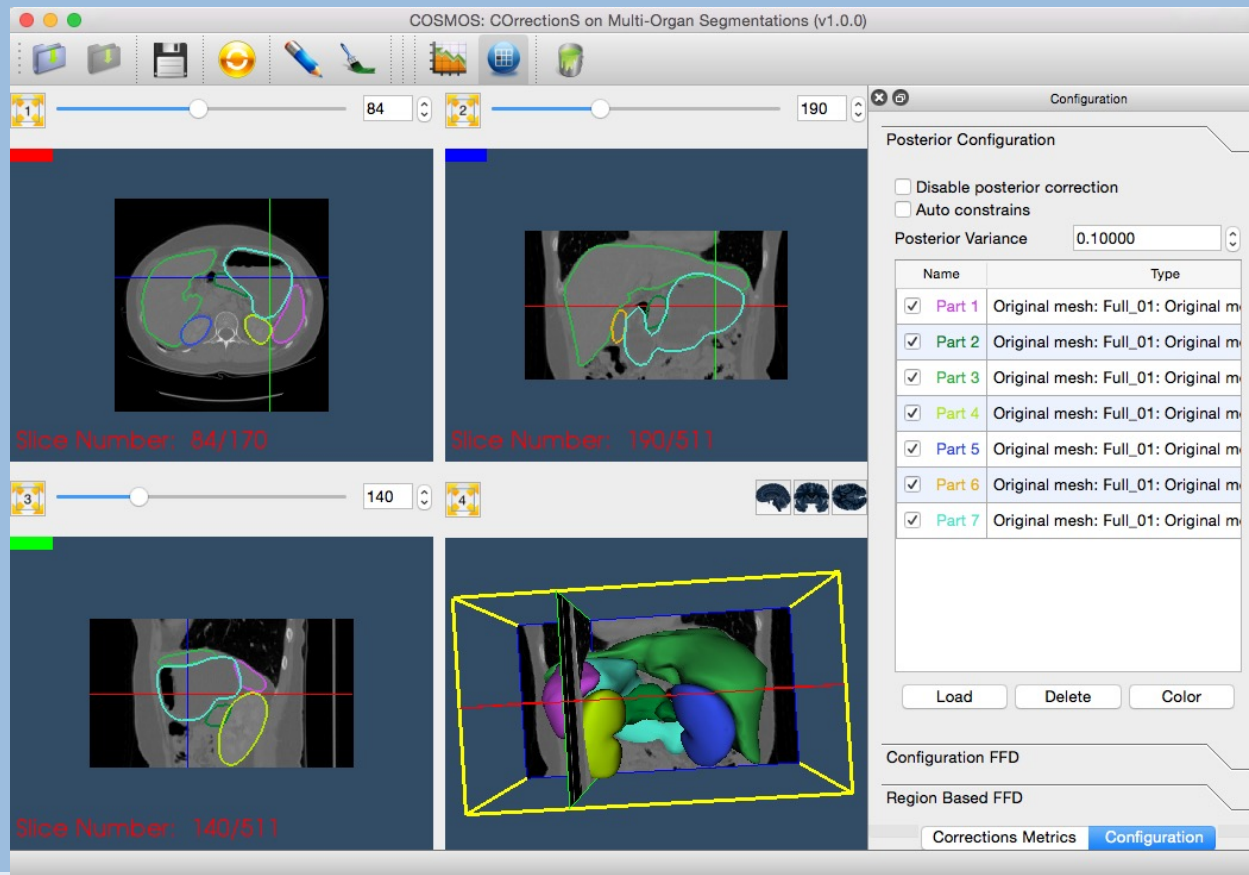
$\mathbf{Q}_g$ and $\mathbf{u}_{ug}$ submatrices of $\mathbf{Q}$ and $\mathbf{u}$

Corrections on MultiOrgan Segmentations :COSMOS



Corrections on MultiOrgan Segmentations :COSMOS

- > Aim: Modify one, correct many!
- > How: Conditional Statistical Shape Models



Combining shape models with supervised learning for bone shape and cortical thickness estimation

Medical Image Analysis 52 (2019) 42–55



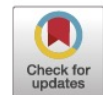
Contents lists available at [ScienceDirect](#)

Medical Image Analysis

journal homepage: www.elsevier.com/locate/media



Supervised learning for bone shape and cortical thickness estimation from CT images for finite element analysis

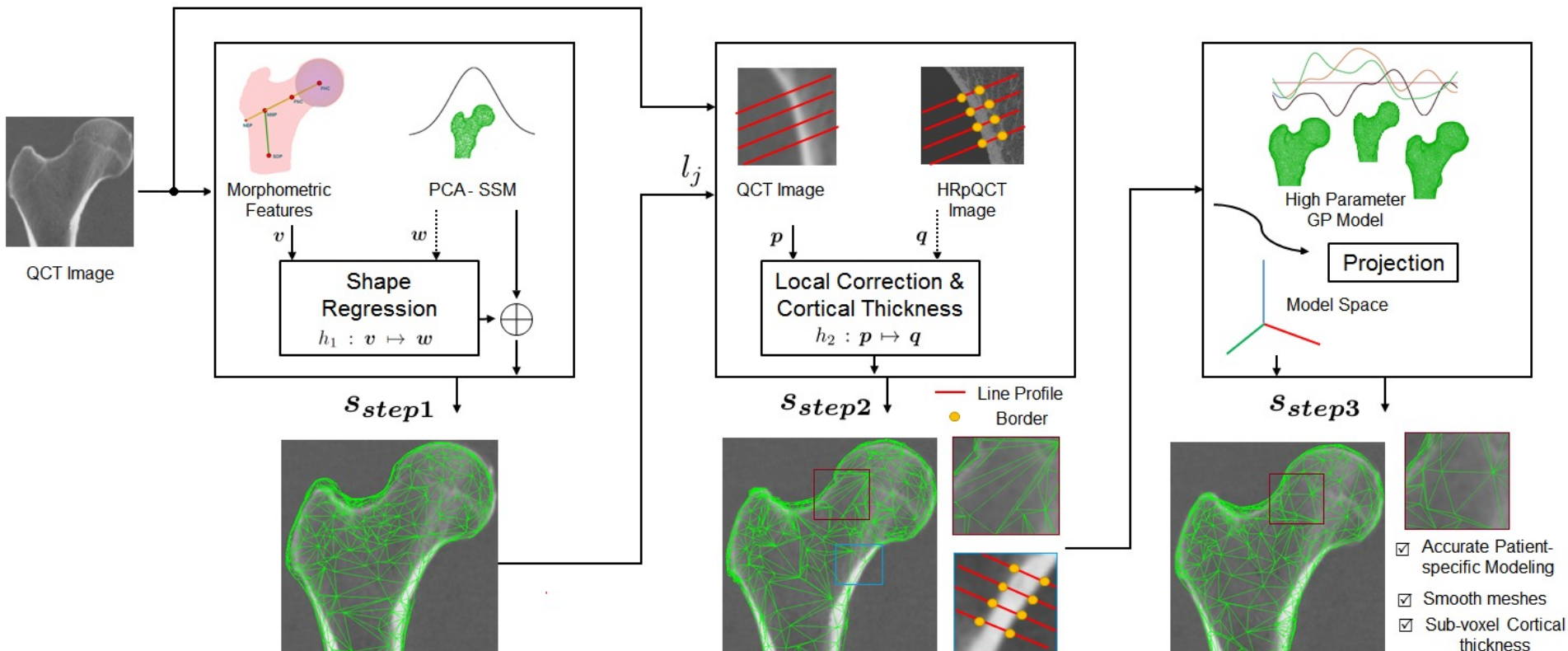


Vimal Chandran^{a,*}, Ghislain Maquer^a, Thomas Gerig^b, Philippe Zysset^a, Mauricio Reyes^{a,*}

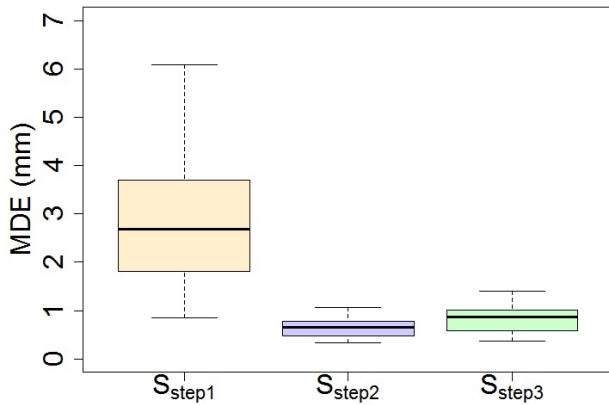
^a Institute of Surgical Technology and Biomechanics, University of Bern, Switzerland

^b Graphics and Vision Research Group, University of Basel, Switzerland

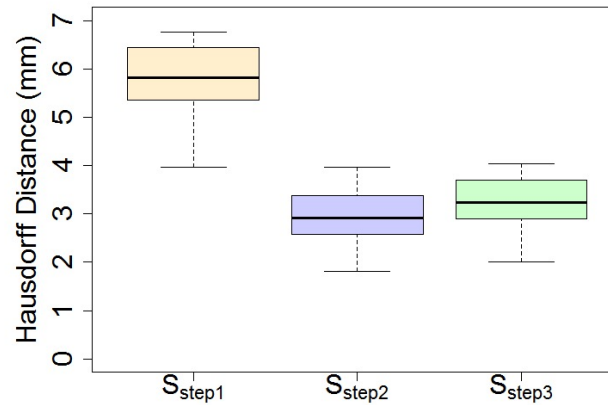
Combining shape models with supervised learning for bone shape and cortical thickness estimation



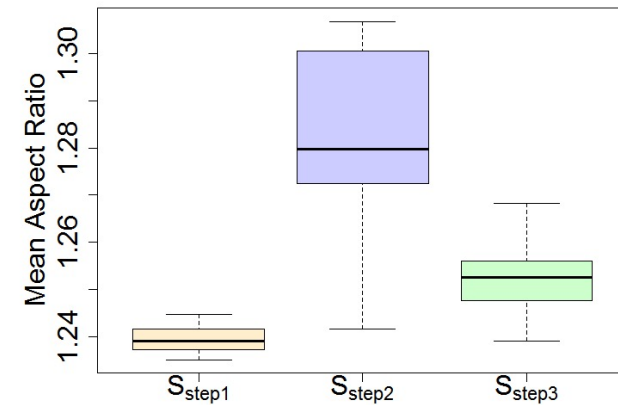
Combining shape models with supervised learning for bone shape and cortical thickness estimation



(a)

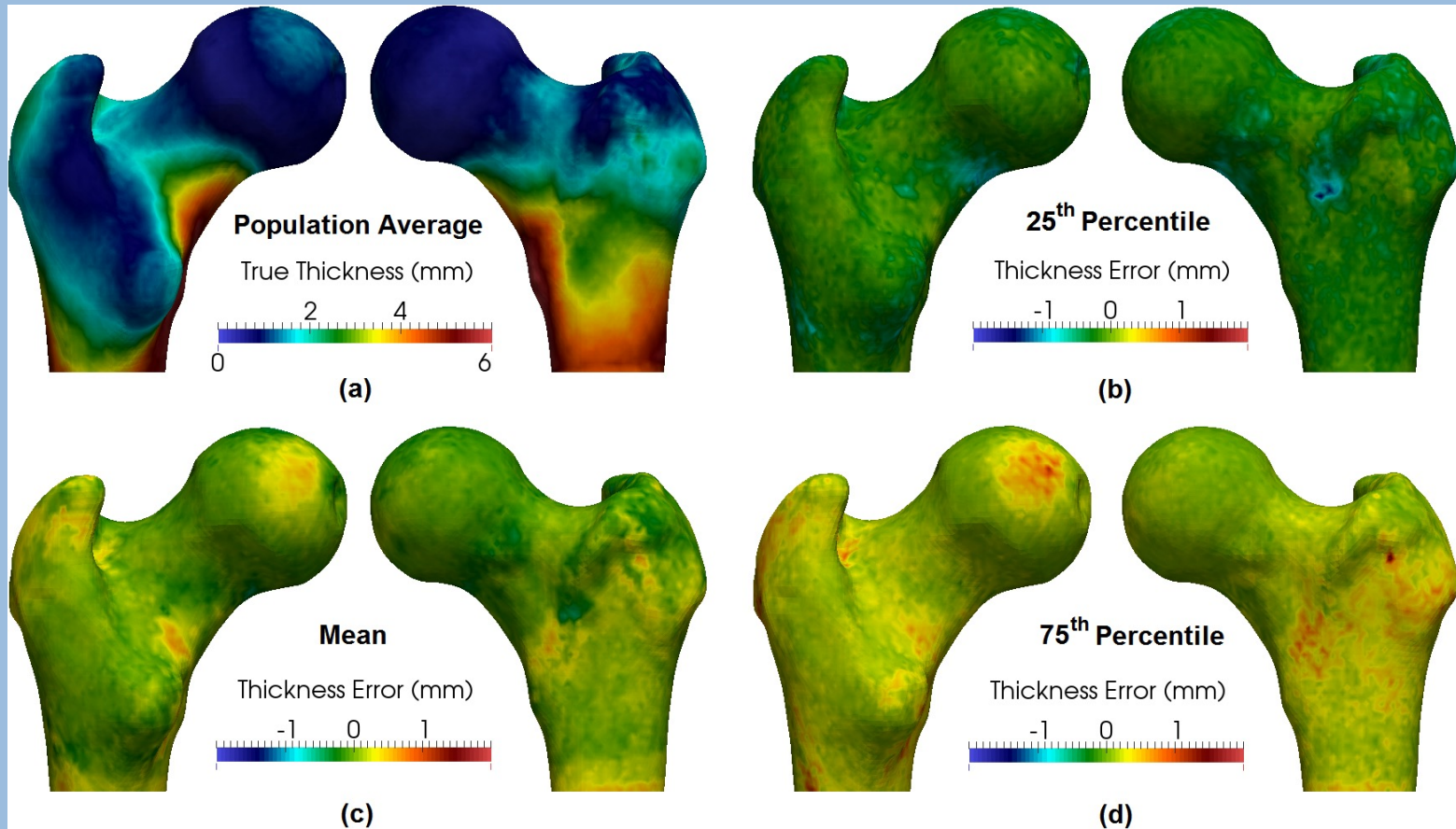


(b)



(c)

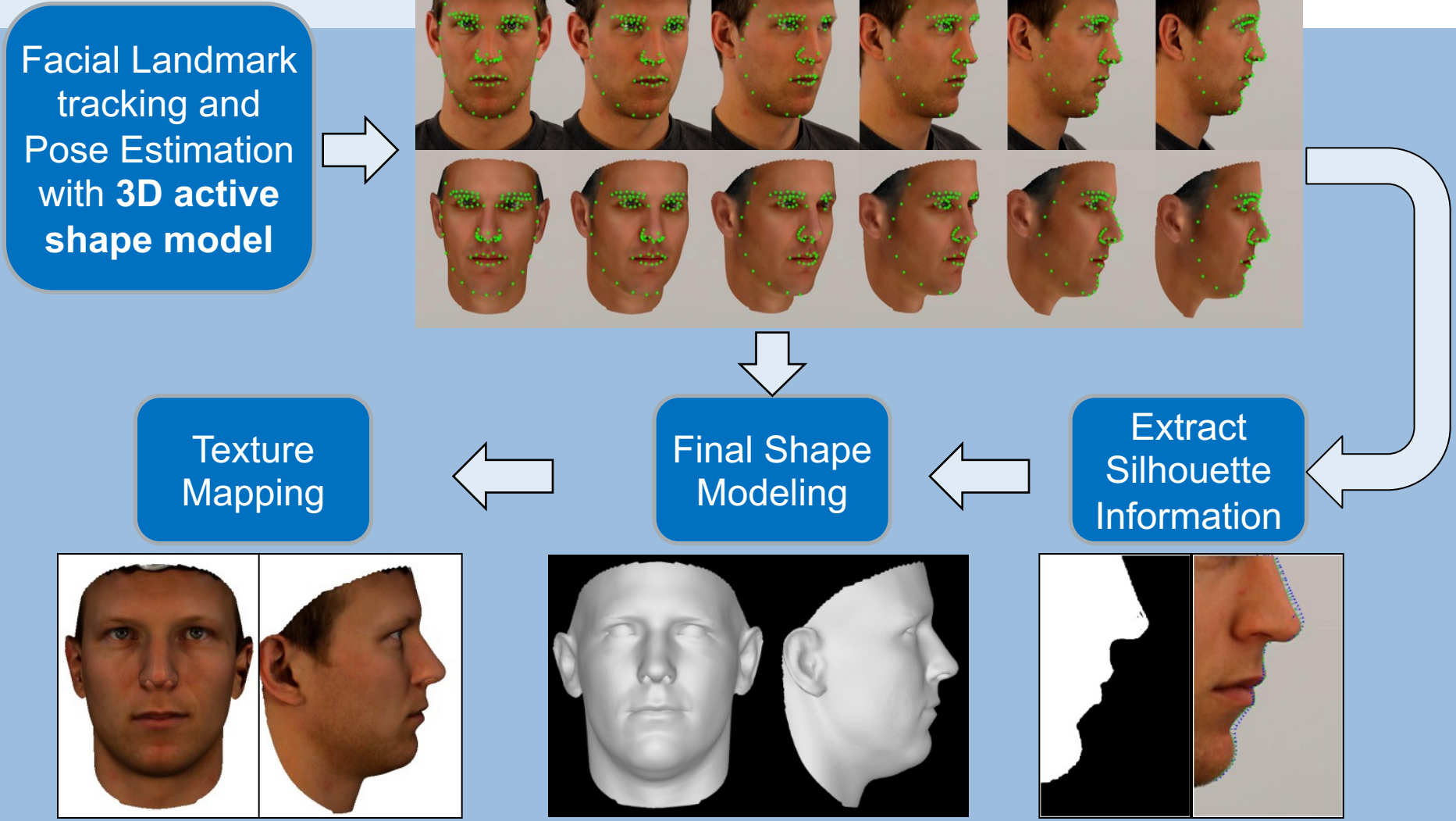
Combining shape models with supervised learning for bone shape and cortical thickness estimation

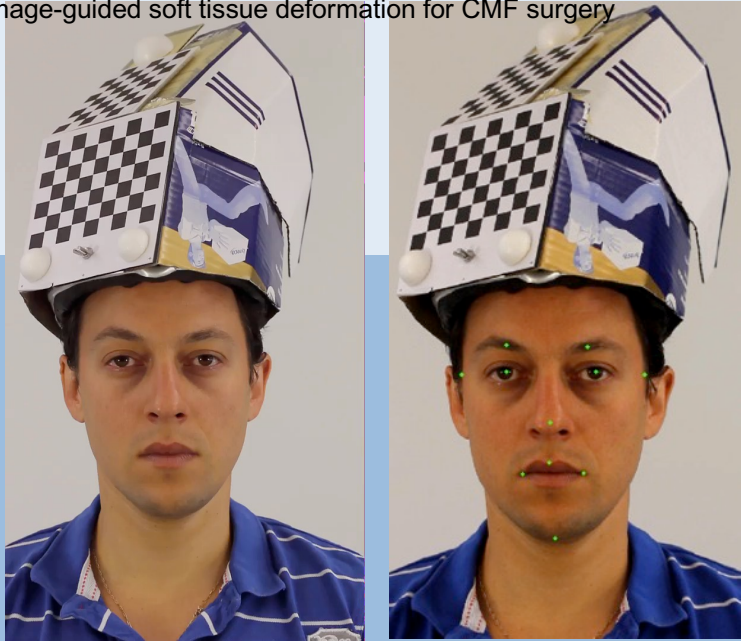


High resolution patient-specific modeling with cortical thickness from clinical CT images

Vimal Chandran, Philippe Zysset, Mauricio Reyes

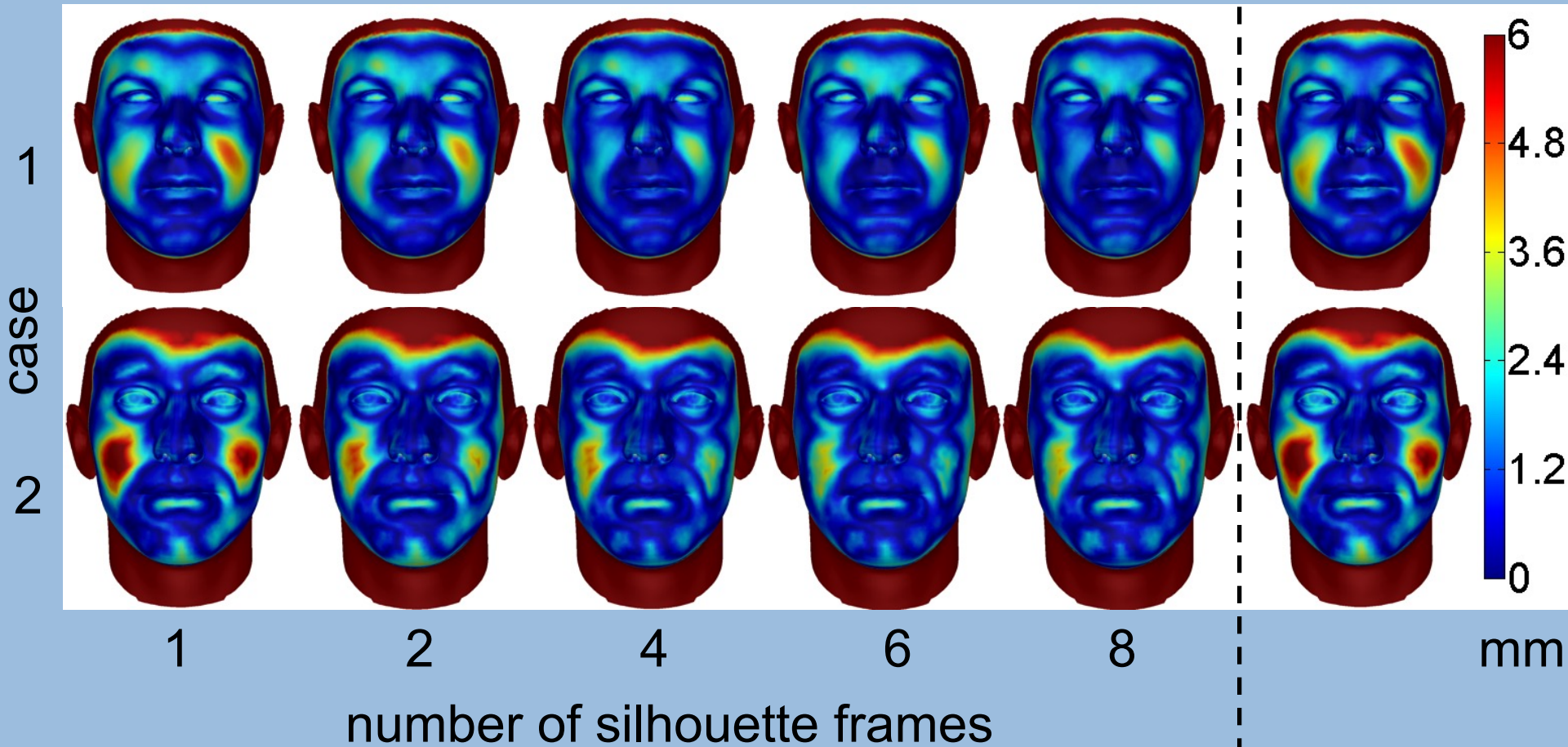
Institute of Surgical Technology and Biomechanics, University of Bern





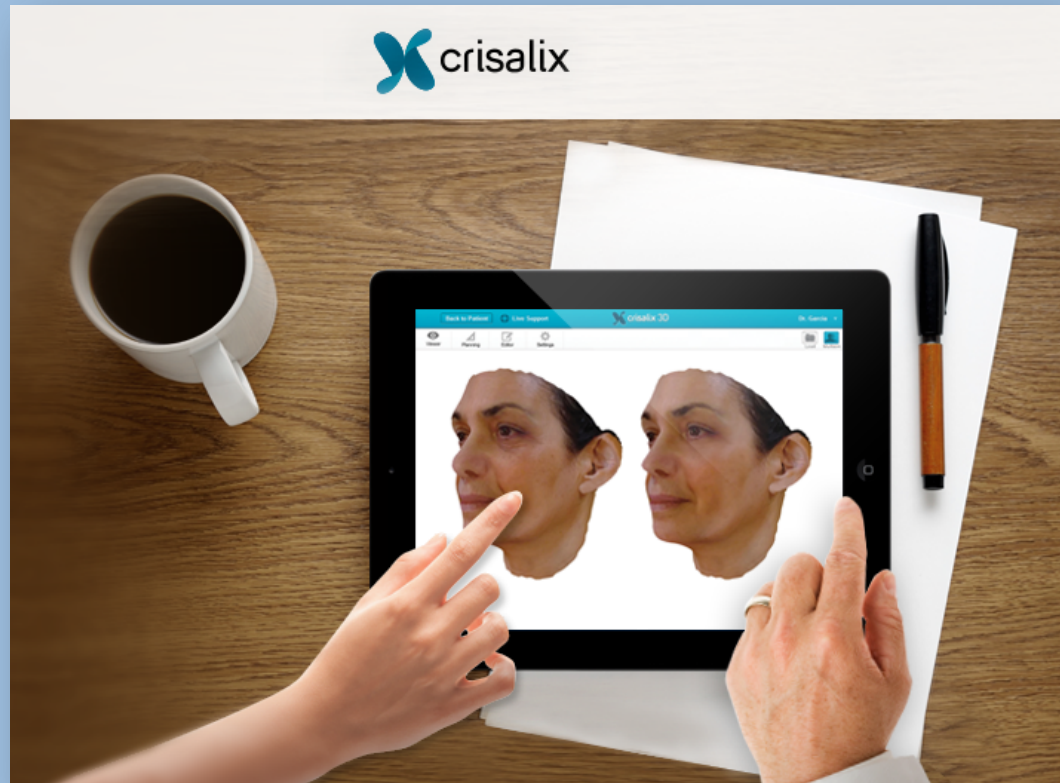
With
silhouette

Without
silhouette



Computer assisted aesthetic face surgery

- > Web-based
- > Improvements over time
- > Data-mining



iPad App

References

- > [1] Cootes T.F., Taylor C.J., Cooper D.H., Graham J., “Active Shape Models – Their training and application” Computer Vision and Image Understanding, 1995
- > [2] Cootes T., Taylor C.J., “Statistical Models of Appearance for Computer Vision”, TechReport University of Manchester, 2004
- > [3] Heimann T., Meinzer H., “Statistical Shape Models for 3D Medical Image Segmentation: A Review”, Medical Image Analysis, 2009